PLANT GREEN INERTIA

PLANT GREEN INERTIA PRIVATE LIMITED
THE WORLD'S FIRST EXPERIENTIAL TRAINING COMPANY
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STUDENT COURSE OUTLINE

WEB DEVELOPMENT | 10-Day Course Outline | 30 Hours

Course overview: Learn how websites are structured, styled, made interactive, tested, and published online.

Day 1 — Introduction to Web Development & HTML

Understand how websites work and learn HTML, the language used to structure webpages. Topics include headings, paragraphs, links, images, buttons, and basic page structure.

Project / application: Create a simple personal profile webpage.

Day 2 — HTML Elements & Forms

Learn lists, tables, input fields, labels, dropdowns, and form elements. Understand how to organize webpage content using semantic sections such as header, navigation, main content, and footer.

Project / application: Create a student registration form.

Day 3 — CSS Fundamentals

Learn how CSS controls the appearance of a webpage, including colours, fonts, backgrounds, borders, spacing, margins, padding, and element sizing.

Project / application: Style the registration page.

Day 4 — Website Layout & Responsive Design

Understand layouts using Flexbox, alignment, navigation bars, cards, and content sections. Learn responsive design so a website can adapt to mobile phones, tablets, and computers.

Project / application: Make a webpage responsive.

Day 5 — JavaScript Basics

Learn variables, data types, operators, conditions, and functions. Understand how JavaScript responds to button clicks, accepts user input, and displays results.

Project / application: Create a calculator or marks calculator.

Day 6 — JavaScript DOM & Form Validation

Learn how JavaScript selects and updates webpage elements. Add required-field checks, simple validation, and helpful success or error messages.

Project / application: Add validation to the student registration form.

Day 7 — Building a Complete Website

Plan and combine multiple sections such as Home, About, Services or Projects, and Contact. Apply consistent styling and connect sections through navigation.

Project / application: Build a portfolio or business landing page.

Day 8 — Website Testing & Improvements

Check the website on different screen sizes, improve readability and usability, optimize images, and use browser developer tools to identify layout or code errors.

Project / application: Test and improve the website.

Day 9 — GitHub & Deployment Preparation

Learn the basics of GitHub repositories and how to upload project files. Prepare the website for online publishing and check file paths, images, and links.

Project / application: Upload the project to GitHub.

Day 10 — Website Publishing & Final Project

Publish a static website using GitHub Pages. Test the published website and understand how to present its design and features.

Project / application: Publish a responsive portfolio or business website.

Tools covered: Visual Studio Code, Google Chrome, HTML, CSS, JavaScript, GitHub and GitHub Pages.

APP DEVELOPMENT | 10-Day Course Outline | 30 Hours

Course overview: Learn mobile app design, user interactions, app logic, data storage, testing, and basic Android app building using MIT App Inventor.

Day 1 — Introduction to Mobile App Development

Understand mobile apps, native apps versus web apps, and the basics of Android and iOS. Explore MIT App Inventor and its Designer and Blocks workspaces.

Project / application: Create a simple welcome screen.

Day 2 — App Screen Design & Components

Learn to add and arrange labels, buttons, text boxes, images, lists, and layouts. Customize colours, fonts, alignment, and screen properties.

Project / application: Design a profile or contact information screen.

Day 3 — Events & User Interaction

Understand event-based programming. Connect buttons to actions, read text entered by users, update screen content, and display messages.

Project / application: Create a greeting app that uses the user's name.

Day 4 — Variables, Conditions & Calculations

Learn variables, simple conditions, mathematical operations, and logic blocks. Use input values to calculate and display results.

Project / application: Build a calculator or marks/grade calculator.

Day 5 — Multiple Screens & Navigation

Learn to organize an app into screens, move between screens, display information, and create simple navigation such as Home and Back.

Project / application: Create a multi-screen quiz app.

Day 6 — Lists & Repeated Data

Learn how to create and display lists of tasks, questions, or expenses, and how to add or remove items using visual programming blocks.

Project / application: Build a basic to-do list app.

Day 7 — Local Data Storage

Understand why apps store information. Use TinyDB to save and retrieve simple values so notes or tasks remain available after closing and reopening the app.

Project / application: Create a notes app with saved entries.

Day 8 — Phone Features & User Experience

Explore a supported phone feature such as sound, camera, or sharing. Learn basic input checks, confirmation messages, error handling, and ways to make an app easier to use.

Project / application: Add a suitable feature to the existing app.

Day 9 — Final App Project Development

Plan the app's screens and user flow, combine interface components with app logic, use stored data where needed, and test the main features.

Project / application: Build an attendance tracker, quiz app, task manager, or expense tracker.

Day 10 — App Testing & Final Project

Test screens and buttons, fix common issues, improve the interface, and prepare a project demonstration. Build an Android APK if supported by the current App Inventor environment.

Project / application: Present a working beginner-level mobile app.

Tool covered: MIT App Inventor — https://appinventor.mit.edu/

Note: This beginner course uses visual programming and focuses on Android-style app development. Professional native Android or iOS development requires a different course.